Vishwa Rajput

AI Engineer

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LinkedIn, GitHub

AI and Data Engineer with 4+ years of experience in software engineering, data analytics, and AI system evaluation. Recently completed an M.Sc. in Computer Science (Artificial Intelligence) with hands-on experience building and evaluating LLM and Vision Language Model based systems. Strong background in prompt engineering, retrieval evaluation, anomaly analysis, root cause investigation, and writing structured technical assessments.

Experience

Working Student Data Engineer • Decathlon • Germany May '25 – Feb '26

- Developed and maintained large-scale data pipelines using Python, SQL, PySpark, Databricks, AWS and Apache Airflow.
- Designed automated data validation, monitoring, and quality-control processes to ensure accuracy, consistency, and completeness of business-critical datasets.
- Implemented Git based CI/CD, automated testing, and data quality workflows to support reliable deployment and operation of production data solutions using AWS-based infrastructure.
- Industrialized AI/ML use cases from development to production, integrating solutions into existing data workflows.

Intern Application Manager • Decathlon • Germany Oct '23 – Apr '25

- Developed marketplace automation solutions using Python, REST APIs, PySpark, and SQL to automate business processes, integrate systems, and improve operational efficiency.
- Analyzed large-scale datasets and created business intelligence dashboards to support decision making.
- Provided application support and troubleshooting, resolving user tickets and technical issues promptly while coordinating with relevant stakeholders to deliver effective solutions.

Software Engineer • Tatvasoft • Ahmedabad, India Nov '20 – Feb '23

- Designed and developed full-stack enterprise applications using C#, .NET Technologies, SQL, REST APIs, MSSQL, Oracle, and MySQL.
- Built backend services and database-driven applications supporting multiple client projects.
- Collaborated in Agile development teams, using Git and GitLab for version control, code collaboration, and software development workflows.

Education

Technical University of Applied Sciences Würzburg-Schweinfurt, Germany | Master Computer Science - Artificial Intelligence Mar '23 – Mar '26
GPA: 2.55/5

Gujarat Technological University, India | Bachelor of Engineering - Computer Engineering Jul '16 – Jun '20
GPA: 8.53/10

Master Thesis

Multi-source Retrieval Using Graph and Vector Hybrid Models in Regulatory Contexts

- Tech stack: Python, LangChain, PDF Parsing, Chroma, Neo4j, HuggingFace, Sentence Transformer, Embedding model, LLM - Qwen 2.5 VL, Ollama, RAGAS, Prompt Engineering
- Designed and implemented Vector RAG, Graph RAG, and Hybrid RAG architectures for complex question-answering scenarios.
- Built end-to-end AI pipelines covering document ingestion, preprocessing, chunking, embedding generation, indexing, retrieval, and response generation.
- Developed evaluation frameworks using RAGAS and multiple retrieval metrics to measure system performance and identify model limitations.
- Conducted systematic experimentation with prompts, embeddings, retrieval strategies, and LLM/VLM configurations to improve response quality.
- Analysed model failures, retrieval errors, and edge cases to understand limitations and improve overall system performance.
- Achieved context recall scores of 84%, 83%, and 85% across Vector RAG, Graph RAG, and Hybrid RAG systems.

Skills

Programming: Python, C#, REST API, SQL
Software Development: Databricks, Git, AWS, GCP, Apache Airflow
Frameworks & Libraries: Tensorflow, OpenCV, LangChain, PySpark, Pytorch, Pandas
Operating Systems: Windows, Linux
Soft skills: Analytical thinking, Problem solving, Communication
Languages: English - Fluent(C1), German - Good(B1), Hindi - Nativ, Gujarati - Nativ
Hobbies: Yoga, Badminton

Projects

Language Check
Decathlon

- Tech stack: Facebook fastText, Python, PySpark, SQL, Apache Airflow, DAG, Databricks, AWS, Git CI/CD
- Developed an automated data pipeline to detect the language of Decathlon marketplace product attributes using Facebook FastText and validate it against the product's target country
- Implemented data processing and validation
- Productionized the pipeline with Apache Airflow DAGs for daily automated execution and monitoring

Message Classification
Decathlon

- Tech stack: Python, PySpark, SQL, Apache Airflow, DAG, Databricks, AWS, Git CI/CD
- Industrialized an automated pipeline to classify multilingual customer queries and route them to the appropriate department
- For improvements, analyzed customer messages to identify keywords and language-specific patterns for different categories and reduced unclassified messages rate from 78% to 65%
- Productionized the classification workflow with daily Airflow scheduling

Auto Rejector
Decathlon

- Tech stack: Python, PySpark, SQL, REST API, Databricks, GCP
- Automated marketplace product validation for 10k+ products/day, approving or rejecting products before online publication.
- Implemented robust validation checks on European marketplace product data stored in Delta Lake
- Developed duplicate-variant detection and integrated validation results with Mirakl through APIs
- 93% reduction in invalid product publications

House Price Prediction | Link

- Tech stack: Python, Machine learning
- Trained a predictive model using XGBoost Regressor with 89% accuracy
- Feature Engineering on dataset, implemented data preprocessing pipelines, visualized data insights

Smart Darts | Link

- Tech stack: OpenCV, Python
- Automated the system of score calculation for Dart game
- Used computer vision methods to pre-process images and perform dart tip detection

Amazon Review Analysis | Link

- Tech stack: Python, NLP, Neural Network
- Trained a deep learning model for sentiment analysis of reviews with 95% accuracy
- Used NLP techniques for tokenization of words and embedding as part of preprocessing

Health questionnaire and temperature check
Tatvasoft

- Tech stack: HTML, CSS, Javascript, jQuery, .Net core MVC, Dapper, MySQL
- Developed a web application to send QR codes in the email based on filled questionnaire by customers
- Implemented CRON job schedule for sending emails